$$\int_{-1}^{1} f(x) P_n(x) = \frac{(-1)^n}{2^n n!} \int_{-1}^{1} (x^2-1)^n \frac{d^n f(x)}{dx^n} \, dx$$

Thus,

$$\int_{-1}^{1} x^{n+2k} P_n(x) = \frac{(-1)^n}{2^n n!} \int_{-1}^{1} (x^2-1)^n \frac{d^n}{dx^n} \left[x^{n+2k} \right] dx$$

Let us simplify $\frac{d^n}{dx^n} \left[x^{n+2k} \right]$

take one derivative $\frac{d^{n-1}}{dx^{n-1}} \left[(n+2k) \, x^{n+2k-1} \right]$

a second one, $\frac{d^{n-2}}{dx^{n-2}} \left[(n+2k)(n-1+2k) \, x^{n+2k-2} \right]$

Do this n-times to exhaust all derivative signs

$$\Rightarrow \quad (n+2k)(n+2k-1)(n+2k-2) \cdots (2k+1) \, x^{2k}$$

[Note this is only true for $k \geq 0$]

This can be written as, $\frac{(n+2k)!}{(2k)(2k-1)\cdots(1)} x^{2k} = \frac{(n+2k)!}{(2k)!} x^{2k}$

Thus, we have

$$\int_{-1}^{1} x^{n+2k} P_n(x) \, dx = \frac{(-1)^n}{2^n n!} \frac{(n+2k)!}{(2k)!} \underbrace{\int_{-1}^{1} (x^2-1)^n x^{2k} \, dx}_{\text{symmetric integral}}$$

$$= \frac{(-1)^n}{2^n n!} \frac{(n+2k)!}{(2k)!} \times 2 \int_{0}^{1} (-1)^n (1-x^2)^n x^{2k} \, dx$$

$$= \frac{(n+2k)!}{2^n\, n!} \times \frac{2}{(2k)!} \underbrace{\int_0^1 (1-x^2)^n\, x^{2k}\, dx}_{\text{Beta function}}$$

$u = x^2$

$du = 2x\,dx$

$= 2u^{1/2}\,dx$

$$= \frac{(n+2k)!}{2^n\, n!} \times \frac{1}{(2k)!} \underbrace{\int_0^1 (1-u)^n\, u^{k-1/2}\, du}_{B(n+1,\, k+1/2)}$$

$$= \frac{(n+2k)!}{2^n\, n!\,(2k)!} \frac{\Gamma(n+1)\,\Gamma(k+1/2)}{\Gamma(n+k+3/2)}$$

Since $\Gamma(n+1) = n!$

$$= \frac{(n+2k)!}{2^n\,(2k)!} \times \frac{\Gamma(k+1/2)}{\Gamma(n+k+3/2)}$$

Second Question: when you compare power series it's better to use Newton binomial expansion in the slides

in your example,

$$\frac{1}{(1+t^2)^{3/2}} = \sum_{n=0}^{\infty} \binom{-3/2}{n} t^{2n}$$

what is left is to simplify $\binom{-3/2}{n}$ using gamma function.